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Information processing apparatus, information processing method, and storage medium

Abstract

To improve reliability of a result of language processing carried out with use of a machine learning model, an information processing apparatus includes at least one processor that carries out: an acquisition process of acquiring a target text; an extraction process of extracting a document related to the target text; a rewriting process of rewriting the target text with use of the document; a generation process of generating a text corresponding to the rewritten target text with use of a machine learning model trained to generate a text based on an input text; and an output process of outputting a result obtained by adding information identifying the document to the generated text.

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Background/Summary

(1) This application is a National Stage Entry of PCT/JP2023/017858 filed on May 12, 2023, the contents of all of which are incorporated herein by reference, in their entirety.

TECHNICAL FIELD

(2) The present invention relates to an information processing apparatus, an information processing method, and a storage medium, all of which carry out language processing on text.

BACKGROUND ART

(3) Techniques that use language models are known. The language models are trained by using text data and are configured to carry out language processing.

(4) Patent Literature 1 discloses a question response apparatus that generates a response to a question sentence based on entities extracted from the question sentence and a value of the perplexity of a language model calculated from the question sentence, the model assuming question sentences of a specific field.

(5) The question response apparatus divides the text of the question sentence into morphemes, which are the smallest units having meanings, that is, words. Then, the apparatus issues a request for keyword search to an external search engine by using nouns included in the words as keywords. Further, the question response apparatus uses a response from the external search engine as the answer to the input question sentence.

CITATION LIST

Patent Literature

Patent Literature 1

(6) Japanese Patent Application Publication Tokukai No. 2015-87796

SUMMARY OF INVENTION

Technical Problem

(7) The question response apparatus disclosed in Patent Literature 1 has not verified whether the response from the external search engine is correct. Therefore, there might be a problem in that the response to the question sentence created by the question answering apparatus is wrong.

(8) An example aspect of the present invention has been made in view of the above problem, and an example object thereof is to provide a technique for improving the reliability of a result of language processing carried out with use of a language model.

Solution to Problem

(9) An information processing apparatus in accordance with an example aspect of the present invention includes at least one processor, the at least one processor carrying out: an acquisition process of acquiring a target text; an extraction process of extracting a document related to the target text; a rewriting process of rewriting the target text with use of the document; a generation process of generating a text corresponding to the rewritten target text with use of a machine learning model trained to generate a text based on an input text; and an output process of outputting a result obtained by adding information identifying the document to the text generated in the generation process.

(10) An information processing method in accordance with an example aspect of the present

invention includes: acquiring, by at least one processor, a target text; extracting, by the at least one processor, a document related to the target text; rewriting, by the at least one processor, the target text with use of the document; generating, by the at least one processor, a text corresponding to the rewritten target text with use of a machine learning model trained to generate a text based on an input text; and outputting, by the at least one processor, a result obtained by adding information identifying the document to the text generated in the generating.

(11) A non-transitory storage medium in accordance with an example aspect of the present invention stores a program for causing a computer to carry out: an acquisition process of acquiring a target text; an extraction process of extracting a document related to the target text; a rewriting process of rewriting the target text with use of the document; a generation process of generating a text corresponding to the rewritten target text with use of a machine learning model trained to generate a text based on an input text; and an output process of outputting a result obtained by adding information identifying the document to the text generated in the generation process.

Advantageous Effects of Invention

(12) According to an example aspect of the present invention, it is possible to improve the reliability of a result of language processing carried out with use of a language model.

Description

BRIEF DESCRIPTION OF DRAWINGS

(1) FIG. 1 is a block diagram illustrating the configuration of an information processing apparatus in accordance with a first example embodiment of the present invention.

(2) FIG. 2 is a flowchart illustrating the flow of an information processing method in accordance with the first example embodiment of the present invention.

(3) FIG. 3 is a block diagram illustrating the configuration of an information processing apparatus in accordance with a second example embodiment of the present invention.

(4) FIG. 4 is a flowchart illustrating the flow of an information processing method in accordance with the second example embodiment of the present invention.

(5) FIG. 5 is a diagram illustrating an example of each process carried out by the information processing apparatus in accordance with the second example embodiment of the present invention.

(6) FIG. 6 is a block diagram illustrating the configuration of an information processing system in accordance with a third example embodiment of the present invention.

(7) FIG. 7 is a table showing an example of documents in the third example embodiment of the present invention.

(8) FIG. 8 is a table showing an example of information on authors of the documents in the third example embodiment of the present invention.

(9) FIG. 9 is a flowchart illustrating the flow of an information processing method in accordance with the third example embodiment of the present invention.

(10) FIG. 10 is a diagram illustrating an example of images displayed on an information processing terminal in accordance with the third example embodiment of the present invention.

(11) FIG. 11 is a block diagram illustrating an example of the hardware configuration of the information processing apparatus in accordance with each of the example embodiments of the present invention.

EXAMPLE EMBODIMENTS

First Example Embodiment

(12) The following description will discuss a first example embodiment of the present invention in detail with reference to the drawings. The present example embodiment is a basic form of example embodiments described later.

(13) (Configuration of Information Processing Apparatus 1)

(14) The following description will discuss the configuration of an information processing apparatus **1** in accordance with the present example embodiment with reference to FIG. **1**. FIG. **1** is a block diagram illustrating the configuration of the information processing apparatus **1** in accordance with the present example embodiment.

(15) As illustrated in FIG. **1**, the information processing apparatus **1** includes an acquisition section **11**, an extraction section **12**, a rewriting section **13**, a generation section **14**, and an output section **15**. The acquisition section **11**, the extraction section **12**, the rewriting section **13**, the generation section **14**, and the output section **15** are configured to realize acquisition means, extraction means, rewriting means, generation means, and output means, respectively, in the present example embodiment.

(16) The acquisition section **11** acquires a target text. The acquisition section **11** provides the acquired target text to the extraction section **12** and the rewriting section **13**.

(17) The extraction section **12** extracts a document related to the target text provided by the acquisition section **11**. the extraction section **12** provides the extracted document to the rewriting section **13**.

(18) The rewriting section **13** rewrites the target text acquired by the acquisition section **11** with use of the document provided by the extraction section **12**. The rewriting section **13** provides the rewritten target text to the generation section **14**.

(19) The generation section **14** generates a text corresponding to the target text rewritten by the rewriting section **13**, with use of a language model trained to generate a text based on an input text. The generation section **14** provides the generated text to the output section **15**.

(20) The output section **15** outputs a result obtained by adding information identifying the document to the text generated by the generation section **14**.

(21) As described in the foregoing, the information processing apparatus **1** in accordance with the present example embodiment employs a configuration including: the acquisition section **11** that acquires a target text; the extraction section **12** that extracts a document related to the target text provided by the acquisition section **11**; the rewriting section **13** that rewrites the target text acquired by the acquisition section **11** with use of the document provided by the extraction section **12**; the generation section **14** that generates a text corresponding to the target text rewritten by the rewriting section **13**, with use of a language model trained to generate a text based on an input text; and the output section **15** that outputs a result obtained by adding information identifying the document to the text generated by the generation section **14**. Therefore, with the information processing apparatus **1** in accordance with the present example embodiment, it is possible to achieve an example advantage of improving the reliability of the result of the language processing carried out with use of the language model.

(22) (Flow of Information Processing Method **S1**)

(23) The following description will discuss the flow of an information processing method **S1** in accordance with the present example embodiment with reference to FIG. **2**. FIG. **2** is a flowchart illustrating the flow of the information processing method **S1** in accordance with the present example embodiment.

(24) (Step **S11**)

(25) In step **S11**, a target text is acquired. The acquisition section **11** provides the acquired target text to the extraction section **12** and the rewriting section **13**.

(26) (Step **S12**)

(27) In step **S12**, the extraction section **12** extracts a document related to the target text provided by the acquisition section **11**. The extraction section **12** supplies the extracted document to the rewriting section **13**.

(28) (Step **S13**)

(29) In step **S13**, the rewriting section **13** rewrites the target text acquired by the acquisition section **11** with use of the document provided by the extraction section **12**. The rewriting section **13**

provides the rewritten target text to the generation section **14**.

(30) (Step **S14**)

(31) In step **S14**, the generation section **14** generates a text corresponding to the target text rewritten by the rewriting section **13**, with use of a language model trained to generate a text based on an input text. The generation section **14** provides the generated text to the output section **15**.

(32) (Step **S15**)

(33) In step **S15**, the output section **15** outputs a result obtained by adding information identifying the document to the text generated by the generation section **14**.

(34) As described in the foregoing, the information processing method **S1** in accordance with the present example embodiment employs a configuration including: acquiring, by the acquisition section **11**, a target text; extracting, by the extraction section **12**, a document related to the target text provided by the acquisition section **11**; rewriting, by the rewriting section **13**, the target text acquired by the acquisition section **11** with use of the document provided by the extraction section **12**; generating, by the generation section **14**, a text corresponding to the target text rewritten by the rewriting section **13**, with use of a language model trained to generate a text based on an input text; and outputting, by the output section **15**, a result obtained by adding information identifying the document to the text generated by the generation section **14**. Thus, with the information processing method **S1** in accordance with the present example embodiment, an example advantage similar to that of the abovementioned information processing apparatus **1** is brought about.

Second Example Embodiment

(35) The following description will discuss a second example embodiment of the present invention in detail with reference to the drawings. The same reference numerals are given to constituent elements which have functions identical to those described in the first example embodiment, and descriptions as to such constituent elements are omitted as appropriate.

(36) (Outline of Information Processing Apparatus **2**)

(37) An information processing apparatus **2** in accordance with the present example embodiment is an apparatus that acquires a target text and outputs a result obtained by carrying out language processing on the target text. Further, to the result outputted from the information processing apparatus **2**, information identifying a document related to the acquired target text is added. The document will be described later. In the following description, the “target text” and the “text” are also referred to as the “string”.

(38) The information processing apparatus **2** carries out language processing with use of a language model **M** trained by using text data. The language model **M** receives, as input, a string (text) and outputs a result (text) obtained by carrying out language processing on the input string. The result outputted from the language model **M** may be a string or alternatively, may be an image. The language processing carried out by the language model **M** is not particularly limited. Examples of the language model **M** include: a process of generating a text based on an input text; text classification; emotion analysis; information extraction; text summarization; text generation; image generation; and question answering.

(39) The following description will discuss the language model **M** in detail. The language model **M** is created by means of learning of the relationship between words in a text (text data), and is a model that generates, from the target string, a related string related to the target string. With use of the language model that has been made to learn statements and texts of various contexts, it is possible to generate the related string of reasonable contents related to the target string.

(40) Examples of the language model **M** may include, but not limited to: large language models (LLMs) such as Bidirectional Encoder Representations from Transformers (BERT), Generative Pre-trained Transformer (GPT), Text-to-Text Transfer Transformer (T5), Robustly optimized BERT approach (RoBERTa), Efficiently Learning an Encoder that Classifies Token Replacements Accurately (ELECTRA); and learning models created by transfer learning or fine tuning with use of a pre-trained model (e.g., Chat Generative Pre-trained Transformer, ChatGPT).

(41) Further, the string generated by the language model M is not limited to a natural language. The language model M may output an artificial language (program source codes or the like) for, for example, a string inputted in a natural language. For example, the language model M receives, as the target string, input of a question “how to acquire data including a specific string from the database?”. For the input, the language model M may output a program source code for carrying out database processing. Alternatively, the language model M may output a natural language corresponding to the string inputted in an artificial language.

(42) The string acquired by the information processing apparatus 2 is not particularly limited, and may be, for example, a string including at least one word and an instruction indicating which language processing is to be carried out. For example, it is assumed that the information processing apparatus 2 acquires a string “what is the main business field of XX Corporation?”. In this case, the string includes: five words, that is, “what is”, “the main”, “business field”, “of”, and “XX Corporation”; and an instruction to respond to the question stating “what is the main business field of XX Corporation?”.

(43) (Configuration of Information Processing Apparatus 2)

(44) The following description will discuss the configuration of the information processing apparatus 2 with reference to FIG. 3. FIG. 3 is a block diagram illustrating the configuration of the information processing apparatus 2 in accordance with the present example embodiment.

(45) As illustrated in FIG. 3, the information processing apparatus 2 includes a control section 20, an input section 25, an output section 26, and a storage section 27.

(46) The input section 25 is an interface for receiving input from a user. As an example, the input section 25 provides information indicating the received user input to the control section 20.

Examples of the input section 25 may include, but not limited to, a mouse, a keyboard, a touch pad, and a microphone.

(47) The output section 26 outputs data. As an example, the output section 26 may be an interface for outputting data to another apparatus connected thereto. As an example of this case, the output section 26 outputs a result outputted from the control section 20 to the another apparatus connected. As another example, the output section 26 may be a display device that displays an image or may be a speaker that outputs sound. When the output section 26 is the display device, the output section 26 displays an image including the result outputted from the control section 20.

(48) The storage section 27 stores data referred to by the control section 20. An example of the data stored in the storage section 27 may be documents. Examples of the storage section 27 may include, but not limited to, a flash memory, a hard disk drive (HDD), a solid state drive (SSD), and a combination thereof.

(49) The storage section 27 also stores the language model M. It should be noted that “the storage section 27 stores the language model M” means that the storage section 27 stores parameters defining the language model M. The language model may be stored in another storage device (e.g., an external server) other than the storage section 27.

(50) The control section 20 controls constituent elements included in the information processing apparatus 2. Further, as illustrated in FIG. 3, the control section 20 includes an acquisition section 11, an extraction section 12, a rewriting section 13, a generation section 14, an output control section 23, a calculation section 21, and a determination section 22. The acquisition section 11, the extraction section 12, the rewriting section 13, the generation section 14, the output control section 23, the calculation section 21, and the determination section 22 are configured to realize acquisition means, extraction means, rewriting means, generation means, output means, calculation means, and determination means, respectively, in the present example embodiment.

(51) The acquisition section 11 acquires data provided by the input section 25. As an example, the acquisition section 11 acquires a string. In the following description, the string acquired by the acquisition section 11 is also referred to as the first string. The acquisition section 11 stores the acquired first string in the storage section 27.

(52) The extraction section **12** extracts a document related to the string. As an example, the extraction section **12** may extract a document related to the first string from the documents stored in the storage section **27**. As another example, the extraction section **12** may extract a document related to the first string from documents stored in the database connected to the information processing apparatus **2** via the network. The extraction section **12** stores the extracted document, in association with the first string, in the storage section **27**.

(53) The method of extracting a document related to the string carried out by the extraction section **12** is not limited. For example, the extraction section **12** may extract a document including the string. The extraction section **12** may extract a document related to the string with use of any existing search engine.

(54) The rewriting section **13** rewrites the string. As an example, the rewriting section **13** rewrites the first string with use of the document extracted by the extraction section **12**. The string rewritten by the rewriting section **13** is also referred to as the second string. The rewriting section **13** stores the second string, in association with the first string, in the storage section **27**.

(55) The method of rewriting the string carried out by the rewriting section **13** is not limited. As an example, the rewriting section **13** may generate the second string by adding a string described in the document to the first string. As another example, the rewriting section **13** may generate the second string by adding, to the first string, both a string described in the document and a string indicating an additional instruction. As another example, the rewriting section **13** may generate the second string by rewriting the first string to obtain a string indicating an additional instruction, and then by adding, to this string, a string described in the document.

(56) The generation section **14** generates a result obtained by carrying out the language processing on the string. As an example, the generation section **14** generates a text, which is a result obtained by carrying out the language processing on the second string. As described above, the generation section **14** may use the language model **M** trained to generate a text based on an input text. That is, the generation section **14** may input the second string into the language model **M** and generate the text outputted from the language model **M** as a result obtained by carrying out the language processing on the second string. The result that is the text generated by the generation section **14** is also referred to as the first result. The generation section **14** stores the generated first result, in association with the second string, in the storage section **27**.

(57) The output control section **23** outputs data to the output section **26**. The output control section **23** corresponds to the output section **15** in the first example embodiment described above. As an example, the output control section **23** outputs a result obtained by adding information identifying the document extracted by the extraction section **12** to the first result generated by the generation section **14**. The output control section **23** may add information identifying some of multiple documents extracted by the extraction section **12**, to the first result generated by the generation section **14**. Examples of the information identifying the document may include the document name, the author name, and the date of publication, and the uniform resource locator (URL) that indicates where the document is stored. The result outputted by the output control section **23** is also referred to as the second result.

(58) The output control section **23** may output, in addition to the second result, the reliability calculated by the calculation section **21** described later. With this configuration, the output control section **23** can present the reliability of the second result to the user.

(59) The output control section **23** may be configured to output the second result as an optimized result, in a case where the determination section **22**, described later, has determined that the reliability calculated by the calculation section **21** exceeds the threshold. With this configuration, the output control section **23** can output the optimized second result with high reliability.

(60) The calculation section **21** calculates the reliability of the result. As an example, the calculation section **21** calculates the reliability of the first result (text) generated by the generation section **14**. The calculation section **21** stores the calculated reliability, in association with the first

result, in the storage section **27**. With this configuration, the calculation section **21** can ascertain to what extent the first result is reliable.

(61) The method of calculating the reliability of the first result carried out by the calculation section **21** is not limited. The calculation section **21** may calculate the reliability of the first result with use of existing techniques such as facticity analysis, which is a technique for analyzing whether an event has actually occurred. As an example, the calculation section **21** may calculate the reliability of the first result with use of both the first result and the document extracted by the extraction section **12**. Examples of a method of calculating the reliability of the first result carried out by the calculation section **21** with use of both the first result and the document extracted by the extraction section **12** may include (a) a method based on inter-word distance, (b) a method based on inter-document distance, or (c) a method based on a learning model.

(62) In a case where the method based on the inter-word distance is used, the calculation section **21** calculates the reliability of the first result based on the inter-word distance between a word included in the first result and a word included in the document. Specifically, the calculation section **21** first calculates the inter-word distance for each combination of a word included in the first result and a word included in the document. The calculation section **21** calculates the reliability of the first result such that the shorter the calculated inter-word distance is, the higher the reliability of the first result is.

(63) In a case where the method based on the inter-document distance is used, the calculation section **21** calculates the reliability of the first result based on the inter-document distance between a sentence included in the first result and a sentence included in the document. Specifically, the calculation section **21** first calculates the inter-document distance between a text included in the first result and a text included in the document. The calculation section **21** calculates the reliability of the first result such that the shorter the calculated inter-document distance is, the higher the reliability of the first result is.

(64) In a case where the method based on the learning model is used, the calculation section **21** uses a learning model that has been learned by machine learning to receive two sentences as input and to output the similarity between the two sentences. In this case, the calculation section **21** inputs, into the learning model, a sentence included in the first result and a sentence included in the document. Then, the calculation section **21** calculates the reliability of the first result such that the higher the degree of the similarity outputted from the learning model is, the higher the reliability of the first result is.

(65) In a case where the extraction section **12** has extracted multiple documents, the calculation section **21** calculates the reliability of the first result for each of the multiple documents. The calculation section **21** may calculate an arithmetic average value of the calculated multiple reliabilities as the reliability of the first result.

(66) As another example, the calculation section **21** may calculate the reliability with use of the first result, the document extracted by the extraction section **12**, and the string acquired by the acquisition section **11**. For example, the calculation section **21** may calculate the reliability, referring to a result obtained by inputting, into the language model **M** used by the generation section **14**, the first result, the document extracted by the extraction section **12**, and the string acquired by the acquisition section **11**. As an example, the calculation section **21** acquires, from the language model **M**, an index indicating how correct the first result obtained with reference to the document is, as a response to the first string. The calculation section **21** calculates the reliability of the first result such that the higher the output result is (e.g., the larger value the index is), the higher the reliability of the first result is. With this configuration, the calculation section **21** can suitably calculate the reliability of the first result with use of the language model **M** that has generated the first result.

(67) The calculation section **21** may input the first result, the document extracted by the extraction section **12**, and the string acquired by the acquisition section **11** multiple times into the language

model M. In this case, the calculation section **21** may score the result outputted from the language model M for each of the multiple times of input (likelihood, positive frequency, etc.), and calculate the reliability of the first result such that the higher the score is, the higher the reliability of the first result is.

(68) Further, instead of the string acquired by the acquisition section **11**, the calculation section **21** may input a string instructing to answer whether the first result is correct or not into the language model M, considering the document identified by the information added by the output control section **23**. In this case also, the calculation section **21** calculates the reliability of the first result such that the more positive the output result is, the higher the reliability of the first result is.

(69) The determination section **22** determines whether the value exceeds a threshold. As an example, the determination section **22** determines whether the reliability calculated by the calculation section **21** exceeds the threshold. The determination section **22** stores the determination result, in association with the reliability, in the storage section **27**.

(70) (Process Carried Out by Information Processing Apparatus 2)

(71) The following description will discuss an information processing method carried out by the information processing apparatus **2**, with reference to FIG. **4**. FIG. **4** is a flowchart illustrating the flow of an information processing method S2 in accordance with the present example embodiment.

(72) Further, an example of each process carried out by the information processing apparatus **2** will be described with reference to FIG. **5**. FIG. **5** is a diagram illustrating an example of each process carried out by the information processing apparatus **2** in accordance with the present example embodiment.

(73) (Step S21)

(74) In step S21, the acquisition section **11** acquires a string (first string). The acquisition section **11** stores the first string in the storage section **27**.

(75) For example, as illustrated in FIG. **5**, the acquisition section **11** acquires a first string str1 indicating “what is the main description of business of XX Corporation?”.

(76) (Step S22)

(77) In step S22, the extraction section **12** extracts a document related to the first string stored in the storage section **27**. The extraction section **12** stores the extracted document, in association with the first string, in the storage section **27**.

(78) For example, as illustrated in FIG. **5**, the extraction section **12** extracts a document doc related to the first string str1 “what is the main description of business of XX Corporation?”. The document doc includes multiple documents (“Related Document 1”, “Related Document 2”, “Related Document 3”, etc.).

(79) (Step S23)

(80) In step S23, the rewriting section **13** rewrites the first string with use of the first string and the document, which are stored in the storage section **27**, to generate a second string. The rewriting section **13** stores the second string, in association with the first string, in the storage section **27**.

(81) For example, as illustrated in FIG. **5**, the rewriting section **13** generates the second string str2 by adding the document doc to the first string str1.

(82) The rewriting section **13** may generate the second string str2 by adding an instruction. For example, the rewriting section **13** may generate the second string str2 by adding a string that instructs to add the document referred to generate the first result, to a section obtained with reference to the document. As an example, as illustrated in FIG. **5**, the rewriting section **13** may add, to the second string str2, a string ins that indicates an additional instruction, that is, “Answer the question in view of the following information. The answer should include citation in the notation of $\{n\}$.” This instruction instructs the language model M, when the result of the language model M includes a string that is obtained with reference to a document “ $\{n\}$ ”, to add the document “ $\{n\}$ ” to the string.

(83) (Step S24)

- (84) In step S24, the generation section 14 generates a string (first result) corresponding to the second string stored in the storage section 27 with use of the language model M trained to generate a text based on the input text. The generation section 14 stores the first result, in association with the second string, in the storage section 27.
- (85) For example, as illustrated in FIG. 5, the generation section 14 inputs the second string str2 into the language model M, to generate a first result res1 outputted from the language model M.
- (86) (Step S25)
- (87) In step S25, the calculation section 21 calculates the reliability of the first result stored in the storage section 27. The calculation section 21 stores the calculated reliability, in association with the first result, in the storage section 27.
- (88) For example, as illustrated in FIG. 5, the calculation section 21 calculates the reliability with use of the first result res1, the document doc, and the first string str1. As an example, the calculation section 21 inputs the first result res1, the document doc, and the first string str1 into the language model M. In this case, the calculation section 21 acquires, from the language model M, an index indicating how correct the first result res1 obtained with reference to the document doc is, as a response to the first string str1. The calculation section 21 calculates the reliability with reference to the index.
- (89) Further, as an example, instead of the document doc, the calculation section 21 may input a document ({1} or {3}) included in the first result res1 illustrated in FIG. 5 into the language model M, to calculate the reliability. Further, for example, instead of the first string str1, the calculation section 21 may input, into the language model M, the string “can it be said that the first result res1 is correct when considering the information of the document indicated by {1}?”, to calculate the reliability.
- (90) (Step S26)
- (91) In step S26, the determination section 22 determines whether the reliability stored in the storage section 27 exceeds the threshold.
- (92) (Step S27)
- (93) In a case where it has been determined that the reliability exceeds the threshold in step S26 (step S26: YES), then, in step S27, the output control section 23 generates a result (second result) obtained by adding information identifying the document stored in the storage section 27 to the first result stored in the storage section 27. The output control section 23 outputs the generated second result as an optimized result.
- (94) For example, as illustrated in FIG. 5, the output control section 23 generates a second result res2 obtained by adding information i_doc identifying the document to the first result res1.
- (95) The output control section 23 may be configured to add information identifying only a document that has been referred to when the language model M generates the first result res1, among the documents doc extracted by the extraction section 12. This configuration can be realized by adding, to the second string str2, the string ins indicating the abovementioned additional instruction, that is, “Answer the question in view of the following information. The answer should include citation in the notation of {n}.”
- (96) For example, it is assumed that the first result res1 includes “{1}” and “{3}” as illustrated in FIG. 5. In this case, the output control section 23 adds the information i_doc identifying the document indicated by “{1}” and the document indicated by “{3}” to the first result res1, to generate the second result res2.
- (97) On the other hand, in a case where it has been determined that the reliability does not exceed the threshold in step S26 (step S26: NO), the information processing apparatus 2 terminates the information processing method S2 without causing the output control section 23 to output the second result.
- (98) (Example Advantage of Information Processing Apparatus 2)
- (99) Thus, the information processing apparatus 2 in accordance with the present example

embodiment carries out language processing, with use of the language model M, on the second string obtained by rewriting the first string with use of the acquired document related to the first string. Therefore, the information processing apparatus **2** can prevent the language model M from carrying out the language processing with reference to a document in which correctness of the contents is unknown, such as a less reliable document, a document that is unclear whether the contents are true or false, and a document whose origin is unknown. Therefore, the information processing apparatus **2** can improve the reliability of the result of the language processing carried out with use of the language model M.

(100) Further, the information processing apparatus **2** outputs a result obtained by adding the information identifying the document related to the first string to the result obtained by the language model M carrying out the language processing on the second string. Therefore, the information processing apparatus **2** can present the user with both the result obtained by carrying out the language processing and the information on the document referred to in the language processing. Therefore, the information processing apparatus **2** can present the user with the reliability of the language processing carried out with use of the language model.

Third Example Embodiment

(101) The following description will discuss a third example embodiment of the present invention in detail with reference to the drawings. The same reference numerals are given to constituent elements which have functions identical to those described in the above example embodiments, and descriptions as to such constituent elements are omitted as appropriate.

(102) (Outline and Configuration of Information Processing System **100**)

(103) The following description will discuss the outline and the configuration of an information processing system **100** in accordance with the present example embodiment with reference to FIG. **6**. FIG. **6** is a block diagram illustrating the configuration of the information processing system **100** in accordance with the present example embodiment.

(104) As illustrated in FIG. **6**, the information processing system **100** includes an information processing apparatus **3** and an information processing terminal **4**. The information processing apparatus **3** and the information processing terminal **4** are connected via a network so that they can communicate with each other. An example of the information processing apparatus **3** may be a stationary computer. Examples of the information processing terminal **4** may include a smartphone, a tablet terminal, and a laptop computer.

(105) In the information processing system **100**, the information processing terminal **4** acquires a target text, and the information processing apparatus **3** carries out language processing on the target text with use of a language model M. Further, the information processing apparatus **3** outputs a result obtained by carrying out the language processing to the information processing terminal **4**. Also in the following description, the “target text” and the “text” are also referred to as the “string”.

(106) As an example, in the information processing system **100**, the user of the information processing terminal **4** creates an account for using the information processing system **100** and registers the basic information (name, age, gender, email address, etc.). When the user inputs a string related to medical counseling into the information processing terminal **4**, the information processing apparatus **3** outputs a response to the medical counseling. Examples of the user input may include symptoms, chronic diseases, and usage conditions of drugs. In response to the medical counseling, the information processing apparatus **3** outputs, in addition to the response, information that supports the response.

(107) (Configuration of Information Processing Apparatus **3**)

(108) As illustrated in FIG. **6**, the information processing apparatus **3** includes a control section **20**, a storage section **37**, and a communication section **38**.

(109) The storage section **37** is identical in configuration and function to the storage section **27** described above. Further, the storage section **37** stores documents and information on authors of the documents.

(110) FIGS. 7 and 8 show examples of the documents and the information on authors stored in the storage section 37. FIG. 7 is a table showing an example of the documents in the present example embodiment. FIG. 8 is a table showing an example of the information on the authors of the documents in the present example embodiment.

(111) As illustrated in FIG. 7, the documents stored in the storage section 37 is those regarding healthcare. As illustrated in FIG. 7, each document stored in the storage section 37 may be stored in such a manner as to be associated with the author or authors, the year of publication, and the summary of the paper.

(112) Further, as illustrated in FIG. 8, information on each author who wrote a document stored in the storage section 37 may be stored in the storage section 37. As illustrated in FIG. 8, as the information on an author who wrote a document, the name of the doctor, the workplace, the career, and the specialty may be stored in the storage section 37.

(113) Thus, the document in the information processing system 100 is a document regarding healthcare and is a document in which the career, specialty, or the like of the author of the document is clear. Therefore, it can be said that the document regarding healthcare in the information processing system 100 is a highly reliable document.

(114) The communication section 38 is an interface for transmitting and receiving data to and from an external apparatus via a network. As an example, the communication section 38 transmits data provided by the control section 20 to the information processing terminal 4 and provides data received from the information processing terminal 4 to the control section 20. Examples of the communication section 38 may include, but not limited to, a communication chip in various communication standards such as Ethernet (registered trademark), Wi-Fi (registered trademark), and radio communications standard for mobile data communications networks, and a USB-compliant connector.

(115) The control section 20 controls constituent elements included in the information processing apparatus 3. Further, as illustrated in FIG. 6, the control section 20 includes an acquisition section 11, an extraction section 12, a rewriting section 13, a generation section 14, an output control section 23, a calculation section 21, and a determination section 22. The acquisition section 11, the extraction section 12, the rewriting section 13, the generation section 14, the output control section 23, the calculation section 21, and the determination section 22 are configured to realize acquisition means, extraction means, rewriting means, generation means, output means, calculation means, and determination means, respectively, in the present example embodiment.

(116) The acquisition section 11 acquires data provided by the communication section 38. As an example, the acquisition section 11 acquires a first string. The acquisition section 11 stores the acquired first string in the storage section 37.

(117) The extraction section 12 extracts a document related to the string. The method of extracting the document carried out by the extraction section 12 is as described above. Further, the extraction section 12 acquires, from the storage section 37, information on an author of the extracted document. The extraction section 12 stores the extracted document and the information on the author of the extracted document in the storage section 37, in association with the first string.

(118) The rewriting section 13 rewrites the string. The method of rewriting the first string and generating the second string, carried out by the rewriting section 13, is as described above.

(119) The generation section 14 generates a result obtained by carrying out the language processing on the string. As described above, the generation section 14 generates a first result obtained by carrying out the language processing on the second string. The method of generating the first result carried out by the generation section 14 is as described above. The generation section 14 stores the generated first result, in association with the second string, in the storage section 37.

(120) The output control section 23 outputs data to the communication section 38. As an example, the output control section 23 outputs a result obtained by adding information identifying the document extracted by the extraction section 12 to the first result generated by the generation

section **14**. Further, the output control section **23** may output a result obtained by adding the information on the author of the document extracted by the extraction section **12**.

(121) Further, as described above, the output control section **23** may output, in addition to the second result, the reliability calculated by the calculation section **21**.

(122) The output control section **23** may be configured to output the second result in a case where the determination section **22** has determined that the reliability calculated by the calculation section **21** exceeds the threshold.

(123) The calculation section **21** calculates the reliability of the result. As an example, the calculation section **21** calculates the reliability of the first result generated by the generation section **14**. The method of calculating the reliability of the first result carried out by the calculation section **21** is as described above. The calculation section **21** stores the calculated reliability, in association with the first result, in the storage section **37**.

(124) The determination section **22** determines whether the value exceeds a threshold. As an example, the determination section **22** determines whether the reliability calculated by the calculation section **21** exceeds the threshold. The determination section **22** stores the determination result, in association with the reliability, in the storage section **37**.

(125) (Configuration of Information Processing Terminal **4**)

(126) As illustrated in FIG. **6**, the information processing terminal **4** includes a control section **40**, an input section **45**, a display section **46**, and a communication section **48**.

(127) The control section **40** controls constituent elements included in the information processing terminal **4**. Further, as illustrated in FIG. **6**, the control section **40** includes an acquisition section **41** and an output control section **42**.

(128) The acquisition section **41** acquires data provided by the input section **45** or the communication section **48**. As an example, the acquisition section **41** acquires a string from the input section **45**. As another example, the acquisition section **41** acquires a second result from the communication section **48**. The acquisition section **41** provides the acquired data to the output control section **42**.

(129) The output control section **42** outputs the data to the display section **46** or the communication section **48**. As an example, the output control section **42** outputs the string acquired by the acquisition section **41** to the communication section **48**. As another example, the output control section **42** outputs the second result acquired by the acquisition section **41** to the display section **46**.

(130) The input section **45** is an interface for receiving data input. Examples of the input section **45** may include a mouse, a keyboard, a touch pad, and a microphone. The input section **45** provides the received data to the control section **40**.

(131) The display section **46** is a device for displaying an image. As an example, the display section **46** displays an image based on the data provided by the control section **40**. Examples of the display section **46** may include a liquid crystal display and an organic electro luminescence (EL) display.

(132) The communication section **48** is an interface for transmitting and receiving data to and from an external apparatus via a network. As an example, the communication section **48** transmits the data provided by the control section **40** to the information processing apparatus **3** and provides data received from the information processing apparatus **3** to the control section **40**. Examples of the communication section **48** may include, but not limited to, a communication chip in various communication standards such as Ethernet, Wi-Fi, and radio communications standard for mobile data communications networks, and a USB-compliant connector.

(133) (Process Carried Out in Information Processing System **100**)

(134) The following description will discuss an information processing method carried out in the information processing system **100**, with reference to FIG. **9**. FIG. **9** is a flowchart illustrating the flow of an information processing method **S100** in accordance with the present example embodiment.

(135) (Step **S101**)

(136) In step **S101**, the acquisition section **41** of the information processing terminal **4** acquires a string (first string). The acquisition section **41** provides the first string to the output control section **42**.

(137) (Step **S102**)

(138) In step **S102**, the output control section **42** outputs the first string provided by the acquisition section **41** to the information processing apparatus **3** via the communication section **48**.

(139) (Step **S103**)

(140) In step **S103**, the acquisition section **11** of the information processing apparatus **3** acquires, via the communication section **38**, the first string outputted from the information processing terminal **4**. The acquisition section **11** stores the first string in the storage section **37**.

(141) (Step **S104**)

(142) In step **S104**, the extraction section **12** extracts a document related to the first string stored in the storage section **37**. Further, the extraction section **12** acquires, from the storage section **37**, information on an author of the extracted document. The extraction section **12** stores the extracted document and the information on the author of the document in the storage section **37**, in association with the first string.

(143) (Step **S105**)

(144) In step **S105**, the rewriting section **13** rewrites the first string with use of the first string and the document, which are stored in the storage section **37**, to generate a second string. The rewriting section **13** stores the second string, in association with the first string, in the storage section **37**.

(145) (Step **S106**)

(146) In step **S106**, the generation section **14** generates a string (first result) corresponding to the second string stored in the storage section **37** with use of the language model **M** trained to generate a text based on the input text. The generation section **14** stores the first result, in association with the second string, in the storage section **37**.

(147) (Step **S107**)

(148) In step **S107**, the calculation section **21** calculates the reliability of the first result stored in the storage section **37**. The calculation section **21** stores the calculated reliability, in association with the first result, in the storage section **37**.

(149) (Step **S108**)

(150) In step **S108**, the determination section **22** determines whether the reliability stored in the storage section **37** exceeds the threshold.

(151) (Step **S109**)

(152) In a case where it has been determined that the reliability exceeds the threshold in step **S108** (step **S108**: YES), then, in step **S109**, the output control section **23** generates a result (second result) obtained by adding information identifying the document stored in the storage section **37** to the first result stored in the storage section **37**, as an optimized result. The output control section **23** may add, to the second result, the result obtained by adding the information on the author of the document stored in the storage section **37**. The output control section **23** outputs, via the communication section **38**, the generated second result to the information processing terminal **4**.

(153) (Step **S110**)

(154) In step **S110**, the acquisition section **41** of the information processing terminal **4** acquires, via the communication section **48**, the second result outputted from the information processing apparatus **3**. The acquisition section **41** provides the acquired second result to the output control section **42**.

(155) (Step **S111**)

(156) In step **S111**, the output control section **42** displays the second result provided by the acquisition section **41** on the display section **46**.

(157) On the other hand, in a case where it has been determined that the reliability does not exceed

the threshold in step **S108** (step **S108**: NO), the information processing system **100** terminates the information processing method **S100** without causing the output control section **23** to output the second result.

(158) Alternatively, the output control section **23** may output, to the information processing terminal **4**, information indicating that no highly reliable result could be generated, via the communication section **38**. In this case, the output control section **42** of the information processing terminal **4** may display on the display section **46** that no reliable result could be generated.

(159) (First Example of Image Displayed by Information Processing Terminal **4**)

(160) An example of the image displayed by the information processing terminal **4** will be described with reference to FIG. **10**. FIG. **10** is a diagram illustrating an example of images displayed on the information processing terminal **4** in accordance with the present example embodiment.

(161) As described above, the output control section **42** of the information processing terminal **4** displays the second result on the display section **46**. As an example, as illustrated at the left diagram of FIG. **10**, the output control section **42** displays the second result res, that is, "Have you also noticed either eye sensitivity to light or slurring of your speech?", in response to the input, "Moderate", which indicates a symptom and is inputted by the user.

(162) As an example, the output control section **42** may display information identifying the document in addition to the second result res. As another example, when the acquisition section **41** acquires information indicating an operation indicating that the user selects the second result res (e.g., an operation performed by touching a portion in the display section **46** at which the second result res is displayed), the output control section **42** may display information identifying the document. Further, as described above, a case where the output control section **23** of the information processing apparatus **3** outputs the second result obtained by adding information on the author of the document, the output control section **42** may display information on the author of the document in addition to the information identifying the document.

Second Example of Image Displayed by Information Processing Terminal **4**

(163) In step **S109** described above, the output control section **23** may output, in addition to the second result, information for communicating with a doctor, to the information processing terminal **4**. As an example, the output control section **23** may refer to the information identifying the document and the information on the author of the document, and then may output the contact information of a doctor in the same professional field as that of the author to the information processing terminal **4**, in addition to the second result.

(164) For example, when the output control section **23** outputs the doctor's contact information to the information processing terminal **4** in addition to the second result in the abovementioned step **S109**, the information processing terminal **4** displays the doctor's contact information on the display section **46** in addition to the second result. In this configuration, when the acquisition section **41** acquires information indicating an operation indicating that the user selects the doctor's contact information (e.g., an operation performed by touching a portion in the display section **46** at which the doctor's contact information is displayed), the output control section **42** establishes contact with the doctor using the contact information.

(165) With this configuration, the information processing terminal **4** enables the user to make a video call with the doctor as illustrated in the central diagram in FIG. **10**.

Third Example of Image Displayed by Information Processing Terminal **4**

(166) In step **S109** described above, the output control section **23** may output, to the information processing terminal **4**, information on an action proposed to the user, in addition to the second result. Examples of actions proposed to the user may include going to a hospital and obtaining a doctor's prescription medicine. The following description will discuss a configuration in which the output control section **23** outputs, to the information processing terminal **4**, information indicating a method of obtaining a doctor's prescription medicine.

(167) In a case where the second result includes information identifying a medicine, the output control section **23** may output, to the information processing terminal **4**, the place of a pharmacy where the medicine is sold, in addition to the second result.

(168) Alternatively, as described above, after the user communicates with the doctor, the doctor may output a prescription for the user to the information processing terminal **4**. When the user obtains the prescription, the information processing terminal **4** may output, in addition to the second result, to the information processing terminal **4**, the place of a pharmacy where the medicine is sold, described in the prescription.

(169) With this configuration, as illustrated in the right diagram in FIG. **10**, the information processing terminal **4** can present the user with a place where the user can purchase the medicine prescribed by the doctor.

First Example Advantage of Information Processing System **100**

(170) Thus, in the information processing system **100** in accordance with the present example embodiment, the information processing apparatus **3** carries out language processing on the second string obtained by rewriting the first string with use of the document related to the first string acquired from the information processing terminal **4**, with use of the language model **M**. Therefore, the information processing apparatus **3** can prevent the language model **M** from carrying out the language processing with reference to a document in which correctness of the contents is unknown, such as a less reliable document, a document that is unclear whether the contents are true or false, and a document whose origin is unknown. Therefore, the information processing apparatus **3** can improve the reliability of the result of the language processing carried out with use of the language model **M**.

(171) Further, in the information processing system **100** in accordance with the present example embodiment, the information processing apparatus **3** outputs the result obtained by adding the information identifying the document related to the first string to the result obtained by the language model **M** carrying out language processing on the second string. Therefore, the information processing apparatus **3** can present the user with both the result obtained by carrying out the language processing and the information on the document referred to in the language processing. Therefore, the information processing apparatus **3** can present the user with the reliability of the language processing carried out with use of the language model.

Second Example Advantage of Information Processing System **100**

(172) In the information processing system **100**, the information processing terminal **4** take a question in medical counseling, and the information processing apparatus **3** generates a response to the medical counseling. The following description will discuss an example advantage achieved in a configuration in which the information processing system **100** outputs a response to medical counseling of the user.

(173) First, in the information processing system **100**, the information processing apparatus **3** generates a response to the medical counseling with use of the language model **M**. In the configuration, the user takes medical counseling via the information processing terminal **4** such as, for example, a smartphone. Therefore, with the information processing system **100**, the user can easily take medical counseling. Further, with the information processing system **100**, the user can easily take medical counseling anytime and anywhere. Further, with the information processing system **100**, for example, the user can receive general information and/or advice about symptoms at a stage before the user receives any diagnosis or treatment.

(174) Further, since the information processing terminal **4** allows the user to ask medical counseling therethrough and outputs a response, the user can take medical counseling without going to the hospital. Further, the information processing system **100** allows the user to obtain a response to medical counseling more quickly than when the user go to the hospital. Further, in the information processing system **100**, response to the medical counseling can be provided without intervention by a doctor, resulting in a decrease in cost of responding to medical counseling.

(175) Further, the language model M used by the information processing apparatus 3 carries out language processing with reference to documents. For example, the information processing apparatus 3 causes the language model M to carried out language processing with reference to documents written by medical specialists, not to documents in which the reliabilities are suspected, such as news articles on the internet. Therefore, the information processing apparatus 3 can present information with higher reliability compared to the case where the conventional language model is used.

Software Implementation Example

(176) Some or all of the functions of the information processing apparatuses 1, 2, and 3 may be realized by hardware such as an integrated circuit (IC chip) or may be alternatively realized by software.

(177) In the latter case, the information processing apparatuses 1, 2, and 3 are each realized by, for example, a computer that executes instructions of a program that is software realizing the functions. FIG. 11 illustrates an example of such a computer (hereinafter referred to as “computer C”). The computer C includes at least one processor C1 and at least one memory C2. The memory C2 stores a program P for causing the computer C to operate as each of the information processing apparatuses 1, 2, and 3. In the computer C, the functions of each of the information processing apparatuses 1, 2, and 3 are realized by the processor C1 reading the program P from the memory C2 and executing the program P.

(178) The processor C1 may be, for example, a central processing unit (CPU), a graphic processing unit (GPU), a digital signal processor (DSP), a micro processing unit (MPU), a floating point number processing unit (FPU), a physics processing unit (PPU), a tensor processing unit (TPU), a quantum processor, a microcontroller, or a combination thereof. The memory C2 may be, for example, a flash memory, a hard disk drive (HDD), a solid state drive (SSD), or a combination thereof.

(179) Note that the computer C may further include a random access memory (RAM) in which the program P is loaded when the program P is executed and/or in which various kinds of data are temporarily stored. The computer C may further include a communication interface for transmitting and receiving data to and from another apparatus. The computer C may further include an input/output interface for connecting the computer C to an input/output apparatus(es) such as a keyboard, a mouse, a display and/or a printer.

(180) The program P can be recorded in a non-transitory tangible storage medium M from which the computer C can read the program P. Such a storage medium M may be, for example, a tape, a disk, a card, a semiconductor memory, a programmable logic circuit, or the like. The computer C can acquire the program P via the storage medium M. The program P can be transmitted via a transmission medium. The transmission medium may be, for example, a communications network, a broadcast wave, or the like. The computer C can acquire the program P also via such a transmission medium.

Additional Remark 1

(181) The present invention is not limited to the foregoing example embodiments, but may be altered in various ways by a skilled person within the scope of the claims. For example, the present invention also encompasses, in its technical scope, any example embodiment derived by appropriately combining technical means disclosed in the foregoing example embodiments.

Additional Remark 2

(182) Some or all of the above example embodiments can be described as below. However, the present invention is not limited to example aspects described below.

Supplementary Note 1

(183) A language processing apparatus including: acquisition means for acquiring a target text; extraction means for extracting a document related to the target text; rewriting means for rewriting the target text with use of the document; generation means for generating a text corresponding to

the rewritten target text with use of a language model trained to generate a text based on an input text; and output means for outputting a result obtained by adding information identifying the document to the text generated by the generation means.

Supplementary Note 2

(184) The information processing apparatus according to Supplementary note 1, further including calculation means for calculating reliability of the text generated by the generation means.

Supplementary Note 3

(185) The information processing apparatus according to Supplementary note 2, further including determination means for determining whether the reliability exceeds a threshold, wherein, in a case where the determination means has determined that the reliability exceeds the threshold, the output means outputs the result obtained by adding the information identifying the document as an optimized result.

Supplementary Note 4

(186) The information processing apparatus according to Supplementary note 2 or 3, wherein the output means outputs the reliability in addition to the result obtained by adding the information identifying the document.

Supplementary Note 5

(187) The information processing apparatus according to Supplementary note 4, wherein the calculation means calculates the reliability, referring to a result obtained by inputting, to the language model, the text generated by the generation means, the document extracted by the extraction means, and the target text acquired by the acquisition means.

Supplementary Note 6

(188) An information processing method including: acquiring, by at least one processor, a target text; extracting, by the at least one processor, a document related to the target text; rewriting, by the at least one processor, the target text with use of the document; generating, by the at least one processor, a text corresponding to the rewritten target text with use of a language model trained to generate a text based on an input text; and outputting, by the at least one processor, a result obtained by adding information identifying the document to the text generated in the generating.

Supplementary Note 7

(189) A program for causing a computer to carry out: an acquisition process of acquiring a target text; an extraction process of extracting a document related to the target text; a rewriting process of rewriting the target text with use of the document; a generation process of generating a text corresponding to the rewritten target text with use of a language model trained to generate a text based on an input text; and an output process of outputting a result obtained by adding information identifying the document to the text generated in the generation process.

Additional Remark 3

(190) Some or all of the above example embodiments can also be described as below.

Supplementary Note 1

(191) An information processing apparatus including at least one processor, the at least one processor carrying out: an acquisition process of acquiring a target text; an extraction process of extracting a document related to the target text; a rewriting process of rewriting the target text with use of the document; a generation process of generating a text corresponding to the rewritten target text with use of a language model trained to generate a text based on an input text; and an output process of outputting a result obtained by adding information identifying the document to the text generated in the generation process.

Supplementary Note 2

(192) The information processing apparatus according to Supplementary note 1, wherein the at least one processor further carrying out a calculation process of calculating reliability of the text generated in the generation process.

Supplementary Note 3

(193) The information processing apparatus according to Supplementary note 2, wherein the at least one processor further carrying out a determination process of determining whether the reliability exceeds a threshold, wherein in a case where it has been determined, in the determination process, that the reliability exceeds the threshold, the result obtained by adding the information identifying the document is outputted as an optimized result in the output process.

Supplementary Note 4

(194) The information processing apparatus according to Supplementary note 2 or 3, wherein, in the output process, the reliability is outputted in addition to the result obtained by adding the information identifying the document.

Supplementary Note 5

(195) The information processing apparatus according to Supplementary note 4, wherein, in the calculation process, the reliability is calculated by referring to a result obtained by inputting, to the language model, the text generated in the generation process, the document extracted in the extraction process, and the target text acquired in the acquisition process.

(196) Note that the information processing apparatus may further include a memory, and this memory may include a program for causing the processor to carry out the acquisition process, the extraction process, the rewriting process, the generation process, the output process, the calculation process, and the determination process. The program may be stored in a computer-readable, non-transitory, tangible storage medium.

REFERENCE SIGNS LIST

(197) **1, 2, 3** Information processing apparatus **11** Acquisition section **12** Extraction section **13** Rewriting section **14** Generation section **15** Output section **21** Calculation section **22** Determination section **23** Output control section **C1** Processor **C2** Memory doc Document res, res2 Second result res1 First result str1 First string str2 Second string

Claims

1. An information processing apparatus comprising: a memory storing instructions; and at least one processor configured to execute the instructions to perform processes comprising: an acquisition process of acquiring a target text; an extraction process of extracting a document related to the target text; a rewriting process of rewriting the target text with use of the document; a generation process of generating a text corresponding to the rewritten target text with use of a machine learning model trained to generate a text based on an input text; a calculation process of calculating reliability of the text generated in the generation process; and an output process of outputting a result obtained by adding information identifying the document to the text generated in the generation process, and of outputting the reliability of the text generated in the generation process, wherein the calculation process includes referring to a result obtained by inputting, to the machine learning model, the text generated in the generation process, the document extracted in the extraction process, and the target text acquired in the acquisition process.

2. The information processing apparatus according to claim 1, wherein the processes further comprise a determination process of determining whether the reliability exceeds a threshold, in a case where the reliability has been determined to exceed the threshold, the output process includes outputting the result obtained by adding the information identifying the document as an optimized result.

3. An information processing method performed by at least one processor and comprising: an acquisition process of acquiring a target text; an extraction process of extracting a document related to the target text; a rewriting process of rewriting the target text with use of the document; a generation process of generating a text corresponding to the rewritten target text with use of a machine learning model trained to generate a text based on an input text; a calculation process of calculating reliability of the text generated in the generation process; and an output process of

outputting a result obtained by adding information identifying the document to the text generated in the generation process, and of outputting the reliability of the text generated in the generation process, wherein the calculation process includes referring to a result obtained by inputting, to the machine learning model, the text generated in the generation process, the document extracted in the extraction process, and the target text acquired in the acquisition process.

4. A non-transitory storage medium storing a program executable by a computer to perform processes comprising: an acquisition process of acquiring a target text; an extraction process of extracting a document related to the target text; a rewriting process of rewriting the target text with use of the document; a generation process of generating a text corresponding to the rewritten target text with use of a machine learning model trained to generate a text based on an input text; a calculation process of calculating reliability of the text generated in the generation process; and an output process of outputting a result obtained by adding information identifying the document to the text generated in the generation process, and of outputting the reliability of the text generated in the generation process, wherein the calculation process includes referring to a result obtained by inputting, to the machine learning model, the text generated in the generation process, the document extracted in the extraction process, and the target text acquired in the acquisition process.
